

Autonomous AML Detection & FinCEN SAR Filing

An intelligent multi-tool compliance solution for financial anomaly scoring and audit-aware reporting.

Proposed Solution

- **Autonomous Multi-Tool Agent**

Parses natural language requests, investigates customer profiles, and extracts suspicious transactions.

- **Hybrid AI Scoring Core**

Combines XGBoost supervised learning with Isolation Forest anomaly detection for zero-day typology coverage.

- **5 Core AML Typologies**

Detects structuring (\$9k-\$9.9k CTR evasion), rapid cash velocity, offshore transfers, and spikes.

- **1-Click Regulatory Reporting**

Auto-generates compliant FinCEN SAR narratives under 31 U.S.C. § 5318(g).

What Makes It Stand Out

- **Query-Aware Audit Summary**

Provides total transparency by showing detected filters, user intent, and tool invocation provenance for every query.

- **Standardized Risk Escalation**

Directly maps risk scores to clear operational actions: REPORT (FinCEN SAR), REVIEW (EDD), or MONITOR.

- **Dual-View Interactive Workbench**

Officers can seamlessly switch between Flagged Customers and Top Suspicious Transactions with 1-click exports.

- **Production Reliability**

Fast real-time inference (<50ms) backed by a 100% passing unit test suite (19/19 tests).

System Architecture & Technical Specifications

End-to-end multi-tool execution workflow and core technical stack.

Agent Architecture Pipeline

1. NLP Intent & Entity Parser

Extracts user query intent, Customer IDs, risk thresholds, and date ranges.

2. AML Feature Engineering

Computes structuring bands (\$9k-\$9.9k), velocity ratios, and offshore risk scores.

3. Hybrid Anomaly Scoring

Evaluates supervised XGBoost + Isolation Forest to output composite risk scores.

4. Risk Classifier & SAR Engine

Assigns risk levels (HIGH/MED/LOW), maps escalation actions, and renders SAR filing.

Key Technical Specifications

• Backend Framework

Python 3.12, FastAPI, Scikit-Learn, XGBoost, Pandas.

• Frontend Workstation

Vanilla JavaScript, HTML5 Canvas, CSS Design Tokens.

• Inference Latency

< 50ms per multi-tool execution path.

• Dataset Generator

Built-in synthetic institutional 5-year AML transaction generator.

• Automated Test Suite

100% pass rate across 19 unit tests (test_sentinel_aml.py).